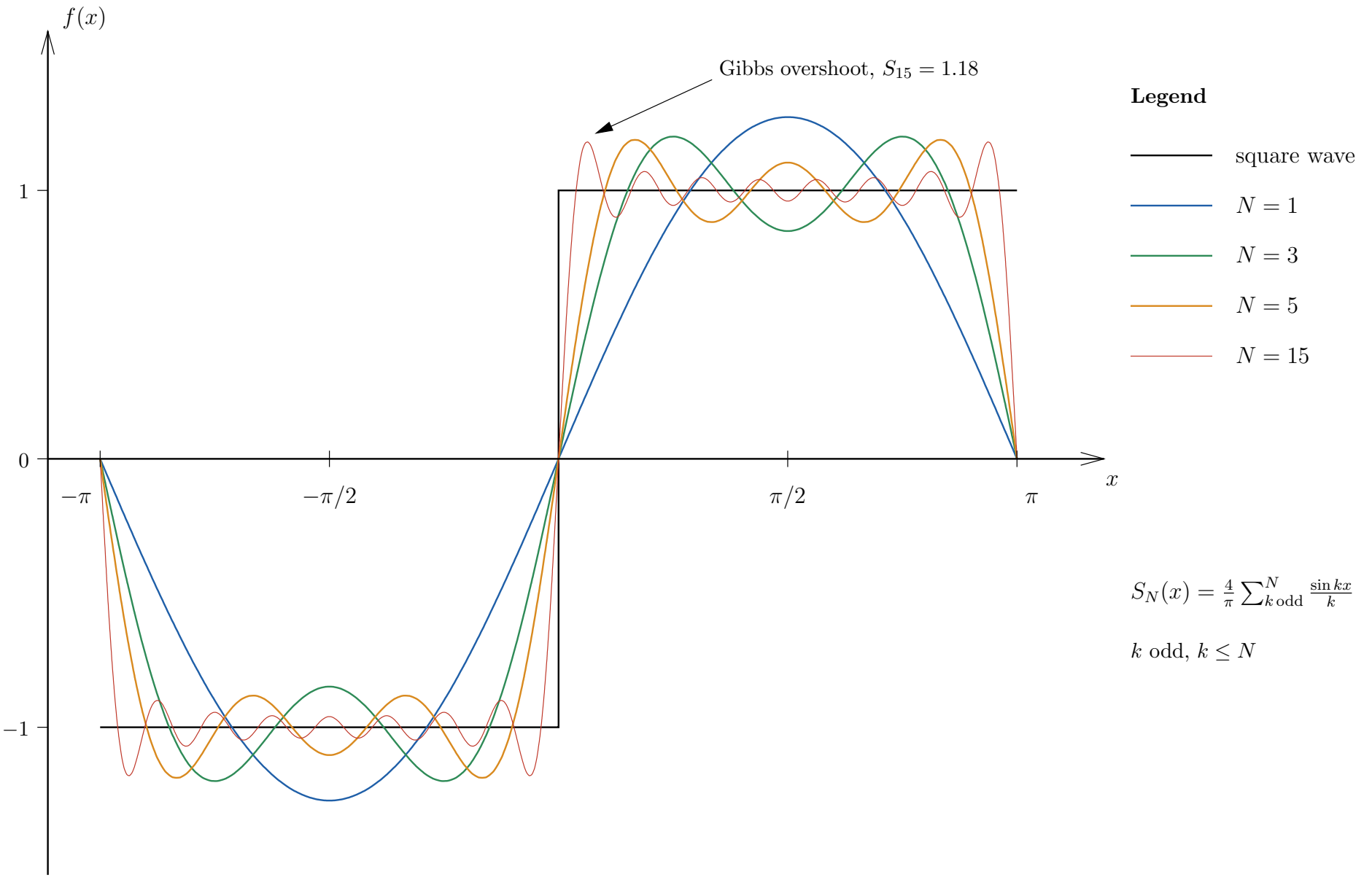
Fourier series of a square wave

Partial sums with 1, 3, 5 and 15 terms; the overshoot at the jump does not vanish (Gibbs)



$$S_N(x) = \tfrac{4}{\pi} \sum_{k\,\text{odd}}^{N} \tfrac{\sin kx}{k}$$

k odd, $k \le N$